

In-app Messaging



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Buyer-seller chat with FCM-driven live updates and inbox management

Category: **Communication** | Server: **REST + FCM push** | Status: **Production**

Listing / Direct Sources	Last-id delta Polling	FCM Push	Text + attach Messages
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1. What it is

WaveMart messaging is a lightweight buyer-seller chat built on REST polling. A conversation can be started from a listing detail screen (one click) or directly from a public profile. The app maintains an always-on inbox with unread counts, last-message snippets and per-conversation polling triggered by FCM pushes so a chat feels live without needing WebSockets.

2. How it works

01 Start from listing 1	POST /api/listings/{id}/message creates a conversation tied to the listing and returns it.
02 Start direct 2	POST /api/users/{id}/message opens a thread not tied to a listing.
03 Inbox 3	GET /api/messages paginates conversations, last message, unread count.
04 Open thread 4	GET /api/messages/{id} returns the full history + the other party's profile + related conversations.
05 Send 5	POST /api/messages/{id} with body (and optional attachment).
06 Live update 6	FCM data message with type=message triggers invalidation of unread count, conversation list and open chat.

3. Capabilities

Capability	What it does
Listing-scoped chat	Tap 'Message seller' on any listing - conversation created in one call.
Direct chat	Tap a user profile - start a thread even if no listing is involved.
Inbox	Sorted by last activity, with unread badge and last snippet preview.

Polling	Lightweight /messages/{id}/fetch with last_message_id for incremental sync.
Push-driven	FCM data messages wake the relevant providers without draining battery.
Suppression	If the user is already in the relevant chat, the local push is suppressed to avoid double-notifying.
Soft delete	DELETE /messages/{id} hides the thread from the user without erasing history for the other side.
Related conversations	Backend groups conversations with the same counter-party to keep the inbox tidy.
Message model	Polymorphic-ready: text body, optional attachment, read_at, soft deletes.

4. Investor relevance

WHY IT MATTERS

Messaging is the trust layer of the marketplace. By keeping buyer-seller communication inside the app, WaveMart captures the conversion data needed for leads, dispute resolution and quality scoring, while delivering a modern UX that pure web classifieds cannot match.

5. Tech surface

Component	Location
Message service	lib/data/services/message_service.dart
Messages screen	lib/presentation/screens/messages/messages_screen.dart (41 KB)
FCM bridge	lib/data/services/fcm_service.dart + LocalNotificationService
API controller	WaveMart/app/Http/Controllers/Api/MessageController.php
Models	WaveMart/app/Models/Conversation.php, Message.php

6. API

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GET /api/messages?page=1 Inbox
GET /api/messages/{id} Open thread (history + other user)
GET /api/messages/{id}/fetch?last_id= Incremental poll
POST /api/messages/{id} { body, attachment_url? }
DELETE /api/messages/{id} Soft delete thread
POST /api/listings/{id}/message Open from listing
POST /api/users/{id}/message Open direct
```